Transformers in language modeling

~~GPT~~ BERT, ResNet

BERT: Bidirectional Encoder Representations from Transformers

It uses unlabeled text for pre-training the deep bidirectional contextual representations.

→ Input	[CLS]	my	dog	is	cute	[SEP]	he	likes	play	##ing	[SEP]
Token Embedings	E_{cls}	E_{my}	E_{dog}	E_{is}	E_{cute}	E_{sep}	E_{he}	E_{likes}	E_{play}	$E_{\#\#ing}$	E_{sep}
Segment Embedings	E_A	E_A	E_A	E_A	E_A	E_A	E_B	E_B	E_B	E_B	E_C
Position Embedding	E_0	E_1	E_2	E_3	E_4	E_5	E_6	E_7	E_8	E_9	E_{10}

BERT input Configuration

Pre-training part

Masked Language Model (MLM)

Randomly mask a percentage of input sequence ... them with a mask token.

tokens and replace

[MASK]

→ 15% mask out rate

My dog is cute. He likes playing.

My dog is cute. He likes [MASK]

→ prob of 80% → token → [MASK]

→ prob of 10% → token → random token

→ prob of 10% → token → unchanged

Cross-entropy loss ⟷ masked tokens

Next Sentence Prediction (NSP)

→ Question Answering (QA)

→ Natural language inference (NLI)

predict if the second sentence should follow the first one.

E[CLS] ⟷ E[SEP] ⟷

50% → < First Sentence > < Second Sentence >

50% → < First Sentence > < Random Sentence >

BERT ↔ MLM, NSP

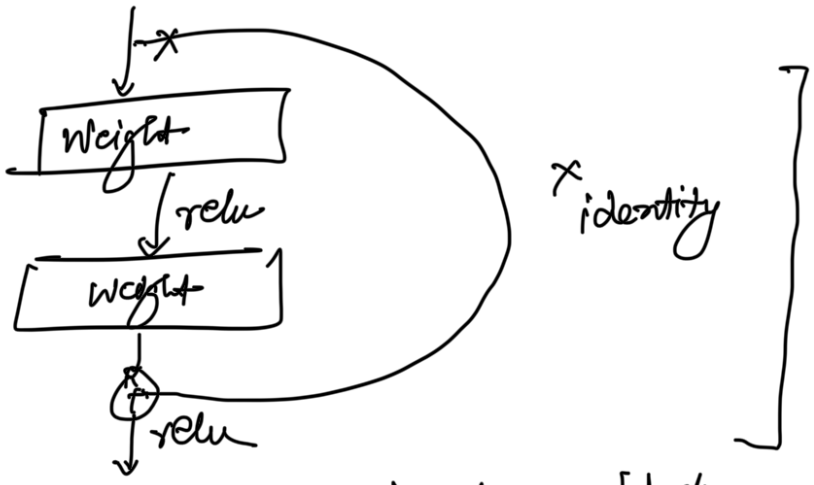
combined loss function

ResNet

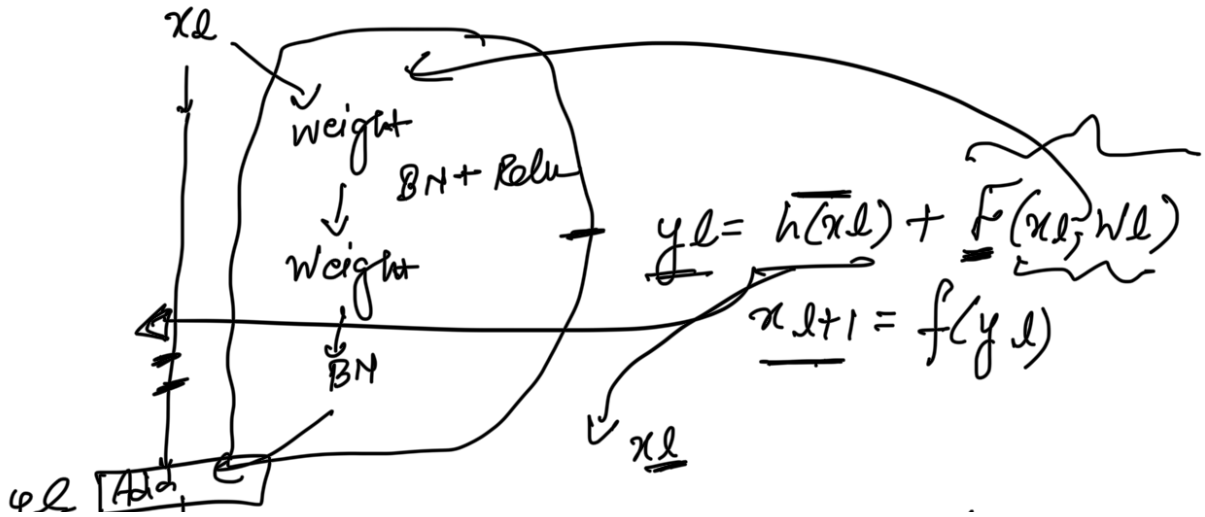
Multimodal sentiment Analysis (BERT / ResNet)

(2015) Deep Residual learning for Image Recognition

Deep NN → Difficult to train



Residual learning: building block



$y_l = h(x_l) + F(x_l, W_l)$

$x_{l+1} = f(y_l)$

$\downarrow$

Relu

$\downarrow$

x_{l+1}

$$x_{l+1} = x_l + F(x_l, w_l)$$

$$\boxed{x_L = x_l + \sum_{i=l}^{L-1} F(x_i, w_i)}$$

BERT $\rightarrow$ strong text representation ability

ResNet $\rightarrow$ image feature extraction